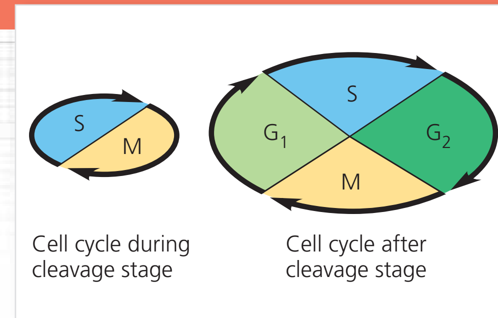
Scientific Skills Exercise

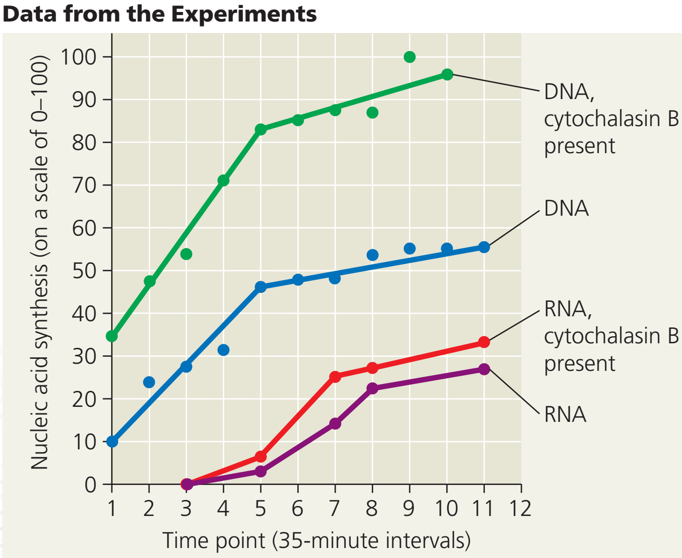
Interpreting a Change in Slope

What Causes the End of Cleavage in a Frog Embryo? For a frog embryo in the cleavage stage, the cell cycle consists mainly of the S (DNA synthesis) and M (mitosis) phases. However, after the 12th cell division, G_1 and G_2 phases appear, and the cells grow, producing proteins and cytoplasmic organelles. What triggers this change?



How the Experiments Were Done Researchers tested the hypothesis that a mechanism for counting cell divisions determines when cleavage ends. They let frog embryos take up radioactively labeled nucleosides, either thymidine (to measure DNA synthesis) or uridine (to measure RNA synthesis). They then repeated the experiments in the presence of cytochalasin B, a chemical that prevents cell division by blocking cleavage furrow formation and cytokinesis.

Data from the Experiments



INTERPRET THE DATA

1. How does the use of particular labeled nucleosides allow independent measurement of DNA and RNA synthesis?
2. Describe the changes in synthesis that occur at the end of cleavage (time point 5 corresponds to cell division 12).
3. Comparing the rate of DNA synthesis with and without cytochalasin B, the researchers hypothesized that the toxin increases diffusion of thymidine into the embryos. Explain their logic.
4. Do the data support the hypothesis that the timing of the end of cleavage depends on counting cell divisions? Explain.
5. In a separate experiment, researchers disrupted the block to polyspermy, generating embryos with seven to ten sperm nuclei. At the end of cleavage, these embryos had the same nucleus-to-cytoplasm ratio as the wild-type embryos, but cleavage ended at the 10th cell division rather than the 12th cell division. What do these results indicate about the timing of the end of cleavage?

Data from J. Newport and M. Kirschner, A major developmental transition in early Xenopus embryos: I. Characterization and timing of cellular changes at the midblastula stage, *Cell* 30:675–686 (1982).

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

CONCEPT CHECK 47.1

1. How does the fertilization envelope form in sea urchins? What is its function?
2. **WHAT IF?** Predict what would happen if Ca^{2+} was injected into an unfertilized sea urchin egg.
3. **MAKE CONNECTIONS** Review Figure 12.16 on cell cycle control. Would you expect MPF (maturation-promoting factor) activity to remain steady during cleavage? Explain.

For suggested answers, see Appendix A.

CONCEPT 47.2

Morphogenesis in animals involves specific changes in cell shape, position, and survival

The cellular and tissue-based processes that are called **morphogenesis** and by which the animal body takes shape occur over the last two stages of embryonic development. During **gastrulation**, a set of cells at or near the surface of the blastula moves to an interior location, cell layers are established, and a primitive digestive tube is formed. Further transformation occurs during **organogenesis**, the formation of organs. We will discuss these two stages in turn.

Gastrulation

Gastrulation is a dramatic reorganization of the hollow blastula into a two-layered or three-layered embryo called a **gastrula**. Cells move during gastrulation, taking up new positions and often acquiring new neighbors. **Figure 47.8** will help you visualize these complex three-dimensional changes. The cell layers produced are collectively called the embryonic **germ layers** (from the Latin *germen*, to sprout or germinate). In the late gastrula, **ectoderm** forms the outer layer and **endoderm** forms the lining of the digestive tract. In a few radially symmetric animals, only these two germ layers form during gastrulation. Such animals are called